



UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering (CSE)
Course Syllabus

1	Course Title	Algorithms Laboratory										
2	Course Code	CSI 228										
3	Trimester and Year	Spring 2023										
4	Pre-requisites	CSI 217: Data Structure, CSI 219: Discrete Mathematics										
5	Credit Hours	1.00										
6	Section	C										
7	Class Hours	Sunday: 11:01 AM – 01:30 PM										
8	Class Room	Lab 4										
9	Instructor’s Name	Mahbub Hossain Raton										
10	Email	mahbub@cse.uiu.ac.bd										
11	Office	319B										
12	Counselling Hours	Sunday	10:00 AM – 11:00 AM									
		Sat and Tue	10:00 AM – 11:30 AM									
13	Text Book	Introduction to Algorithms (3 rd edition) by Cormen, Leiserson, Rivest and Stein										
14	Course Contents (approved by UGC)	Laboratory works based on CSI 227.										
15	Course Outcomes (COs)	<table><tr><th>COs</th><th>Description</th></tr><tr><td>CO1</td><td>Implement correct algorithms to handle large datasets efficiently.</td></tr><tr><td>CO2</td><td>Analyze worst-case running times of algorithms using asymptotic analysis.</td></tr><tr><td>CO3</td><td>Describe different algorithm paradigms and explain when algorithmic design situations call for them. Recite algorithms that employ these paradigms. Synthesize such algorithms. Derive and solve problems describing the performance of the algorithms.</td></tr></table>			COs	Description	CO1	Implement correct algorithms to handle large datasets efficiently.	CO2	Analyze worst-case running times of algorithms using asymptotic analysis.	CO3	Describe different algorithm paradigms and explain when algorithmic design situations call for them. Recite algorithms that employ these paradigms. Synthesize such algorithms. Derive and solve problems describing the performance of the algorithms.
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16	Teaching Methods	Lecture, Case Studies.										
17	CO with Assessment Methods	<table><tr><th>CO</th><th>Assessment Method</th><th>(%)</th></tr><tr><td>-</td><td>Attendance</td><td>10</td></tr></table>			CO	Assessment Method	(%)	-	Attendance	10		
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CO1, CO3	Class Performance	35%
	Presentation	10%
CO1	Assignments	25%
CO2, CO3	Final Quiz	20%

18 Mapping of COs and Program outcomes

COs	Program Outcomes(POs)											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1			C									
CO2		C										
CO3			C									

19 Lab Outline

Class	Topics/Assignments	COs	Lab Outcomes/Activities
Lab1	Practice 1: Review of Recursive Functions	CO1	Lecture, Graded practice
Lab2	Exam 1: Review of Recursive Functions	CO1	Exam
Lab3	Practice 2: Divide-and-Conquer	CO1, CO3	Lecture, Graded practice
Lab4	Exam 2: Divide-and-Conquer Assignment 1	CO1, CO3	Exam; Lecture
Lab5	Practice 3: Greedy Algorithms	CO1, CO3	Lecture, Graded practice
Lab6	Assignment 2: Greedy Algorithms; Practice 4: Dynamic Programming	CO1, CO3	Lecture, Graded practice
MIDTERM WEEK			
Lab7	Exam 3: Dynamic Programming	CO1, CO3	Exam
Lab8	Practice 5: Disjoint-Sets Forests	CO1, CO3	Lecture, Graded practice
Lab9	Exam 4: Disjoint-Sets Forests; Minimum Spanning Trees	CO1, CO3	Exam
Lab10	Practice 6: Single-Source Shortest Paths	CO1, CO3	Lecture, Graded practice
Lab11	Exam 5: Single-Source Shortest Paths Assignment 3	CO1, CO3	Exam
Lab12	Practice 7: String Matching	CO1, CO3	Lecture, Graded practice

